

A Correlational Study on the Relationship of Body Mass Index (BMI) and Dietary Intake of Elementary Students in PC Hill Elementary School

Abstract

This study investigated the relationship between Body Mass Index (BMI) and dietary intake among elementary students at PC Hill Elementary School, addressing the growing concern of childhood obesity. The research aimed to assess the demographic profiles of students, their dietary habits, and how these factors related to their BMI. Utilizing a quantitative, descriptive, correlational approach, data were gathered through the Food Frequency Questionnaire (FFQ) and the students’ demographic records of height and weight. The findings revealed that a substantial percentage of students fell within the normal BMI range, while others were classified as overweight or underweight. Dietary intake patterns indicated a preference for staple foods like rice, alongside occasional consumption of snacks. Analysis showed significant relationships between certain foods and BMI, particularly highlighting calorie-dense items such as pancakes, corn, and chicken as positively influencing BMI. Conversely, increased consumption of fruits, especially bananas, was linked to lower BMI scores. The study emphasized the need for improved nutritional education for students, parents, and school canteen administrators to promote healthier dietary choices and combat the rising trend of childhood obesity.

Background of the Study



Escalating
at an alarming rate, childhood obesity has become a critical public health crisis, particularly in developing nations. (Bhadoria et al., 2015).

1 in 10
Filipino children and teens are overweight, according to a 2019 DOST-FNRI survey (Reyes, 2022).

390 million
children and adolescents aged 5–19 were overweight in 2022 (WHO).

Shifting environments
push children toward sugary, high-fat, and calorie-dense food choices.

Significance



Students



Teachers

Objectives



To determine the demographic profiles of grades 4, 5, and 6 elementary students at PC Hill Elementary School in terms of: a. Height b. Weight



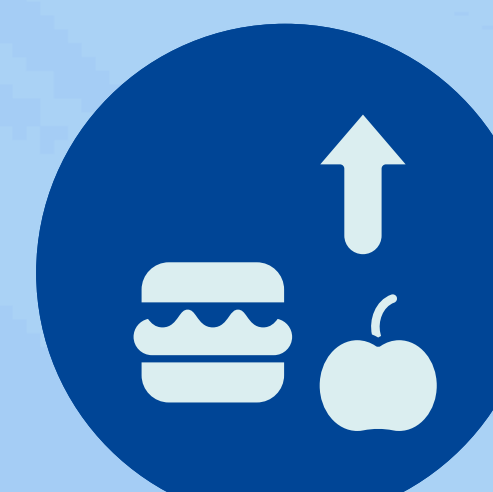
To assess the average Body Mass Index (BMI) of the elementary students.



To evaluate the frequency of the elementary students' dietary intake.



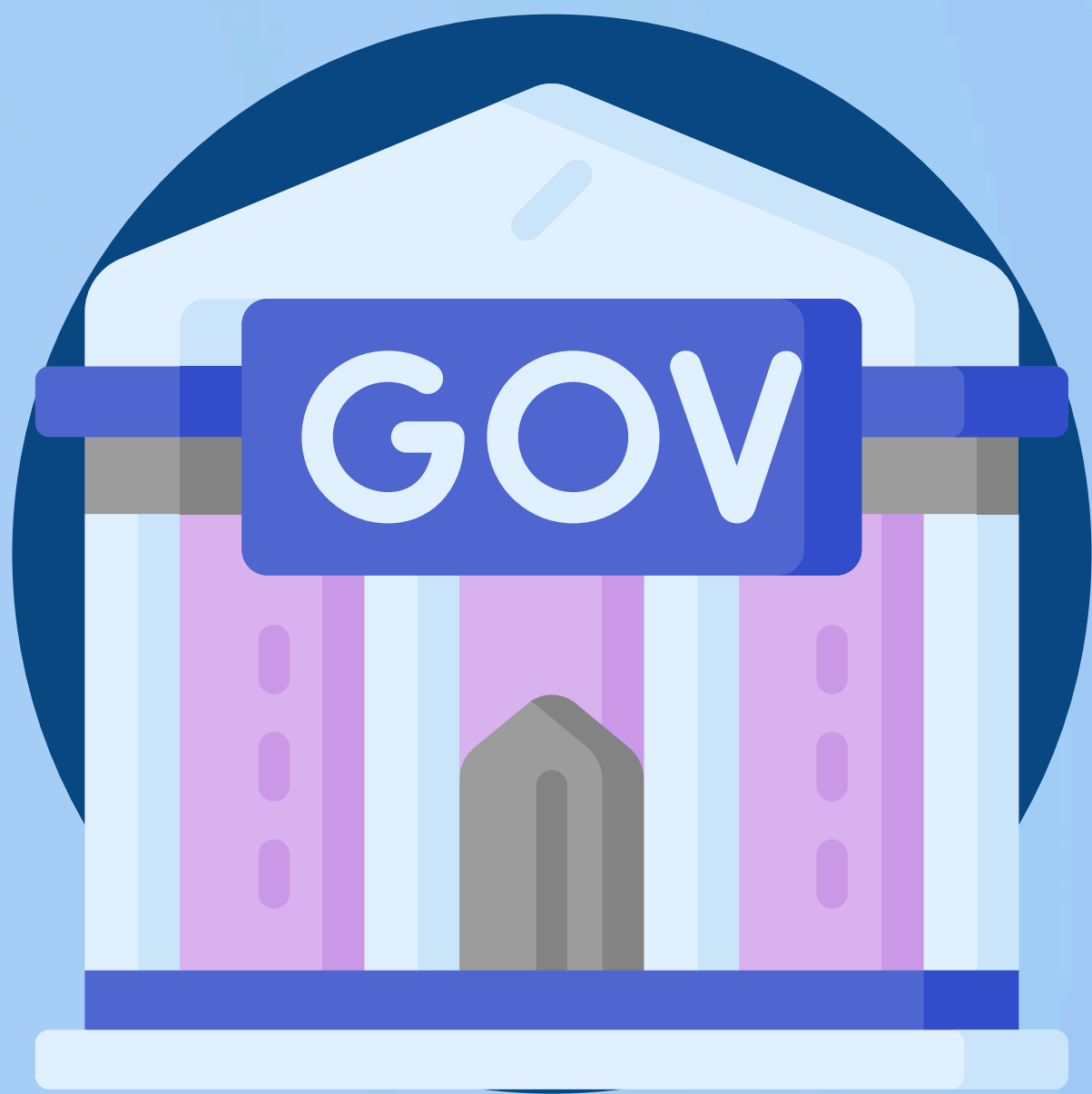
To measure the amount of dietary intake among the elementary students.



To determine if there is a significant relationship between the Body Mass Index (BMI) and dietary intake of the elementary students.



Parents



National & Local Government



Future Researchers

Results and Discussion

Height Range (cm)	Frequency	Percentage
120–129	9	15.2%
130–139	19	31.7%
140–149	29	48.4%
150–159	3	5.0%

Most respondents belonged to the 140–149 cm height range (48.4%) and 30–39 kg weight range (38.5%), indicating that the majority of students had average height and weight measurements.

Weight Range (kg)	Frequency	Percentage
10-19	1	1.7%
20-29	13	22.3%
30-39	23	38.5%
40-49	17	28.3%

BMI Classification	Frequency	Percentage
Underweight	3	5.0%
Normal	40	66.7%
Overweight	17	28.3%

The majority of respondents had a normal BMI classification (66.7%), while 28.3% were overweight and only 5.0% were underweight.

Food Item	Most Common Frequency	Most Common Scale
Rice	Daily	Medium
Potato	Never	Small
Noodles	Occasionally	Small
Pancake	Sometimes	Medium
Banana	Sometimes	Medium
Milk	Daily	Small
Chicken	Sometimes/Daily	Medium
Chips	Sometimes	Small

Rice was the most frequently consumed food and commonly eaten in medium portions, while potatoes, tomatoes, and cabbage were among the least preferred food items.

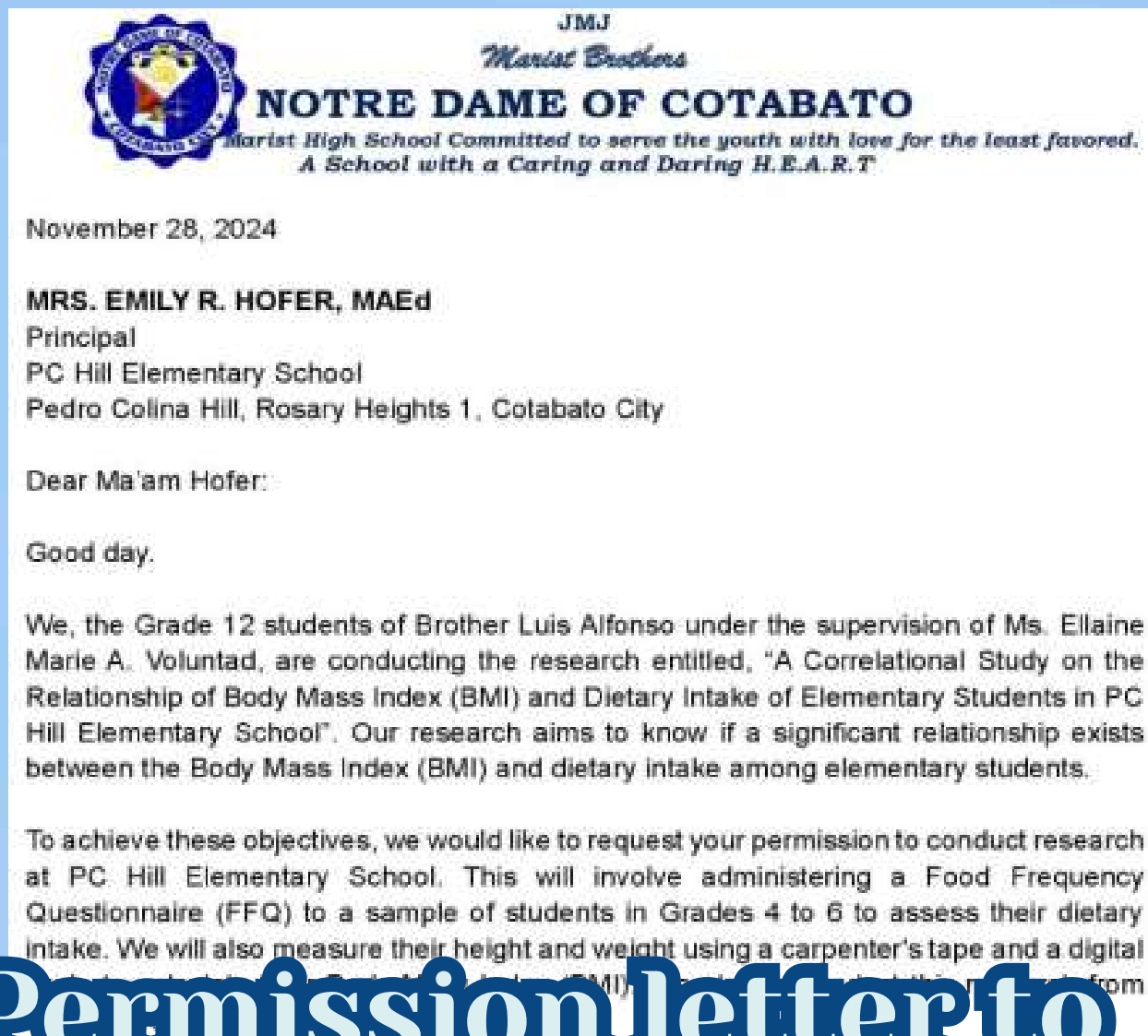
Significant Food Items	p-value	Interpretation
Potato	.035	Significant
Cabbage	.016	Significant
Camote Tops	.048	Significant
Hotdogs	.001	Significant

Potato, cabbage, camote tops, and hotdogs showed significant relationships with BMI based on frequency of consumption.

Significant Food Items	p-value	Interpretation
Pancake	.035	Significant
Corn	.015	Significant
Chicken	.014	Significant

Pancake, corn, and chicken intake significantly influenced BMI scores based on the amount of food consumed.

Methodology



Permission letter to conduct the study outside the campus.



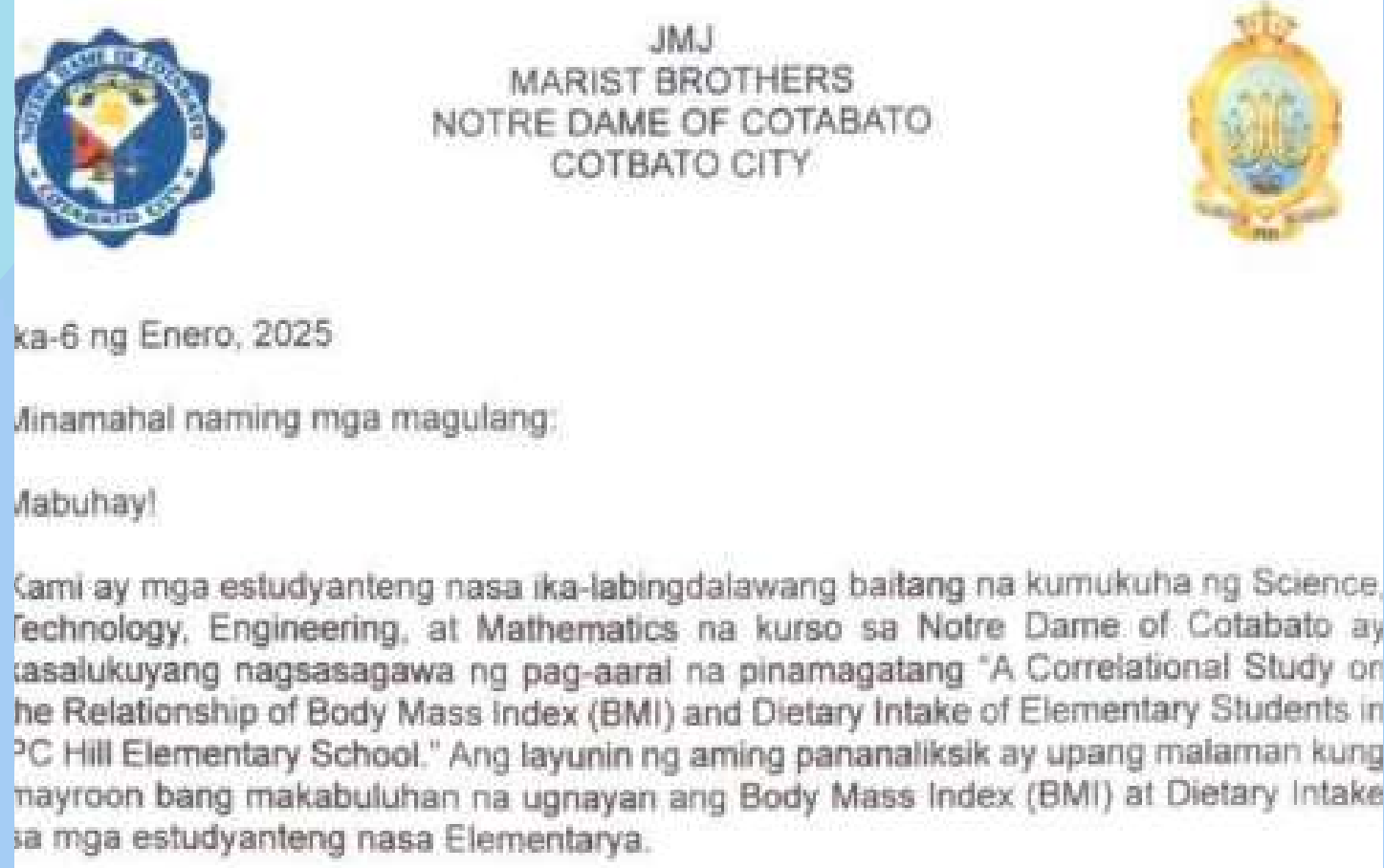
Finalization of survey questionnaire.



Data Collection



Validation of questionnaire to a nutritionist-dietician.



Sending out of parents' consent.



Consultation with the Statistician for Data Analysis and Interpretation

Conclusion

The study highlighted how a lack of nutritional knowledge leads to poor food choices, emphasizing the need for better education, diverse meal options, and community initiatives like feeding programs. Crucially, the statistical data proved a direct link between calorie-dense, processed foods (e.g., hotdogs) and higher BMIs, while high-fiber choices like fruits correlated with a lower BMI.

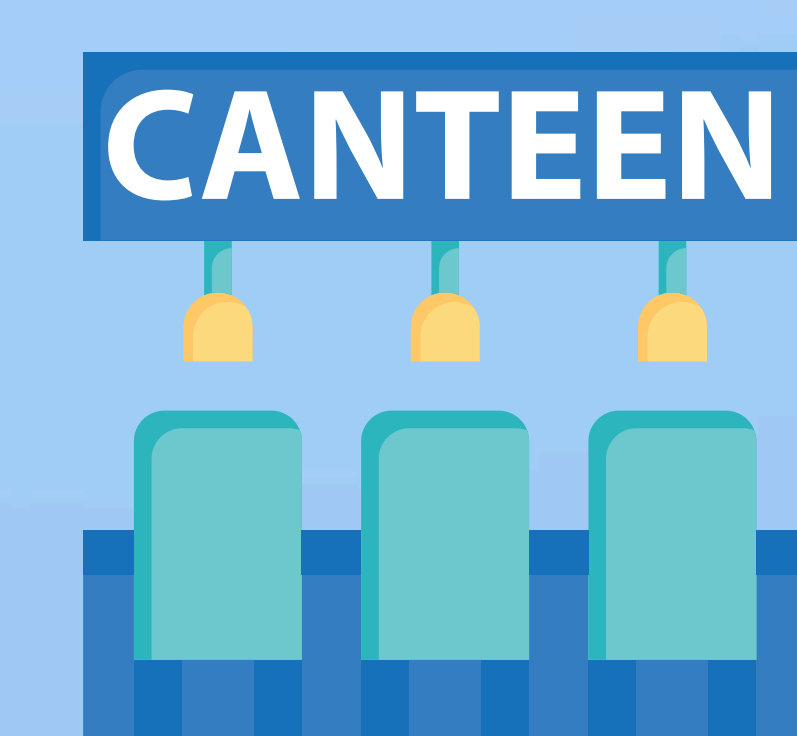
Recommendations



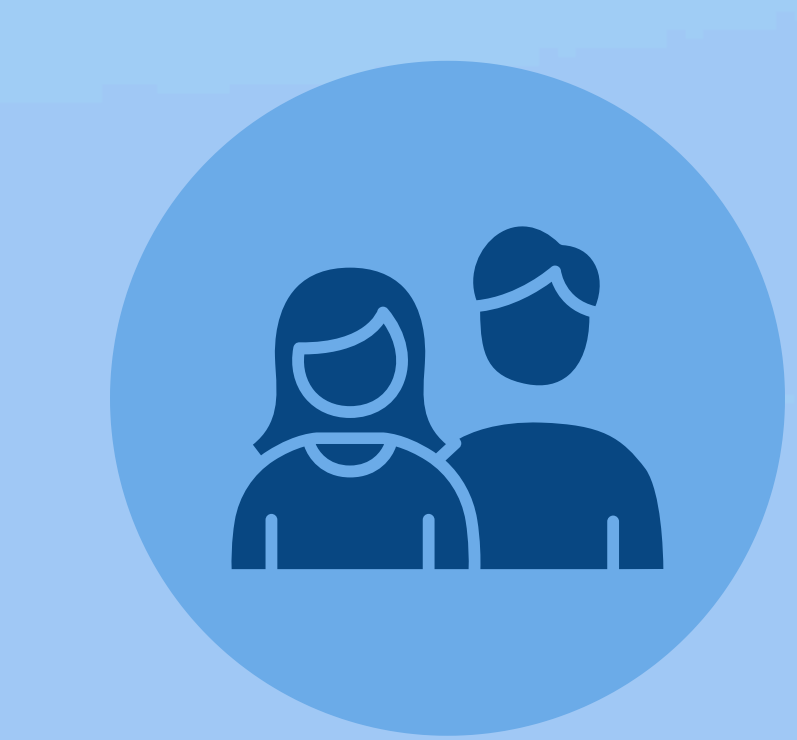
Enhance nutrition education to promote healthier diets and counter rising childhood obesity.



Expand the sample size across multiple districts to create a broader demographic profile.



Swap ultra-processed foods (like instant noodles and hotdogs) for nutrient-dense local staples.



Manage food portions and serve high-fiber options (like bananas and cabbage) to lower obesity risks.

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